Neural-field design of broadband Rayleigh-wave carpet cloaks under microstructure realisability constraints

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Abstract

Transformation elasticity provides exact material distributions for elastodynamic cloaks, but the required stiffness tensors generally violate the minor symmetries of classical Cauchy elasticity and are difficult to realize using conventional materials. Existing approaches restore these symmetries by modifying or symmetrizing the transformed tensor, producing only an approximate cloak. Here we formulate 2D Rayleigh-wave carpet-cloak design as a PDE-constrained optimization problem, using a coordinate-based neural network and a differentiable finite-element solver to optimize symmetric stiffness and density fields by minimizing wave-field distortion. We consider both single-frequency and broadband optimization, with the broadband model trained simultaneously over multiple frequencies.

Physical realizability is addressed using a database of homogenized microstructures through conditional diffusion, neural-field inverse design, and nearest-neighbour selection. FEM simulations show that the unconstrained design approaches ideal transformation-based cloaking, while the best realizable design retains approximately 97% of this performance, providing a direct route from continuum optimization to microstructured Rayleigh-wave cloaks.

Keywords: elastic cloaking, Rayleigh waves, transformation elasticity, neural field, implicit neural representation, metamaterials

1. Introduction

Transformation elasticity is a systematic way to steer elastic waves around an object. A coordinate map $\mathbf{x} = \Xi(\mathbf{X})$ opens up room for the object to hide

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in, and rewriting the elastodynamic equations in the mapped coordinates prescribes a position-dependent material in a surrounding coating that guides waves past the object as if it were not there [1, 2]. It is the mechanical counterpart of transformation optics [3]. The "carpet" variant, which hides a defect on an otherwise flat free surface [4], is the configuration most accessible to experiment [5, 6].

The coating material follows from the Brun–Guenneau–Movchan (BGM) push-forward, $c^{\mathrm{eff}}_{ijkl} = J^{-1}\, C_{IjKl}\, F_{iI}\, F_{kK}$ and $\rho^{\mathrm{eff}} = \rho\, J^{-1}$, where $\mathbf{F} = \nabla_X \mathbf{x}$ is the transformation gradient and $J = \det \mathbf{F}$ [1, 2]. The resulting tensor keeps the major symmetry of classical elasticity but breaks the minor symmetries, so the ideal cloak is a polar (Cosserat) medium that lies outside the class of ordinary Cauchy materials [7, 8]. Turning this exact but non-realisable prescription into a cloak that can be built from common materials is the problem we address.

We focus on Rayleigh waves, which travel along the free surface of a half-space in 2D. Their exact cloak is the polar tensor described above, and approaches to realising it with ordinary materials fall into three families: symmetrising it into a genuine Cauchy material, at the price of only approximate cloaking [9, 2]; reproducing the broken minor symmetry with a purpose-built lattice whose ordinary spring-and-mass elements, endowed with distributed restoring torques, assemble into a macroscopically polar (Cosserat) medium [10]; or selecting orthotropic Cauchy cells from a precomputed database, as done for quasi-static mechanical cloaks [11]. For Rayleigh waves specifically, Chatzopoulos et al. [12] compared triangular and semi-circular carpet cloaks and symmetrised their tensors by the arithmetic mean, reporting the symmetrised quadratic semi-circular design as the best performing.

The triangular geometry is, however, especially attractive. Its transformation is a shear with a constant gradient in each half of the cloak, so the ideal tensor is homogeneous there – two mirror-image phases – and non-singular; being frequency-independent, it cloaks exactly at every frequency. Our approach to find a design inside the Cauchy class of ordinary materials is a differentiable pipeline built on a JAX-FEM solver [13] that evaluates a wave-field cloaking objective, with gradients flowing end-to-end through the FEM adjoint. We first use it to search for the best single realisable material – the orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ that most closely reproduce the ideal response, rather than fixing them by symmetrisation. To improve on it while staying realisable, we then discretise the cloak into a regular grid of cells and let a coordinate-based neural field assign each cell its own Cauchy moduli and density; this per-cell freedom compensates for the broken minor symmetry better than any single homogeneous material can. We optimise at a single frequency and then across a target band. This improves substantially on the symmetrised designs of Chatzopoulos et al. [12]. The neural field acts as an implicit material representation, of the kind used for scene reconstruction in computer vision [14, 15, 16]. Reparameterising a design field by a neural network and optimising it through a differentiable solver has been used extensively in structural topology optimisation, with convolutional [17] and coordinate-based, mesh-free networks [18, 19]. Those methods minimise static compliance over a scalar density field; here the network instead outputs an anisotropic stiffness tensor and density and is trained against a frequency-domain elastodynamic cloaking objective, so as to approximate the non-symmetric transformation-elasticity material.

We then take the design one step further, to an explicit realisation. Each cell is filled with a two-phase (solid/void) microstructure drawn from a database of homogenised unit cells [20]; the cells share a single common base material and differ only in their internal geometry, so the whole cloak can be manufactured from one constituent by varying the pattern from cell to cell. To our knowledge no previous study has demonstrated a Rayleigh-wave elastodynamic cloak realised this way from a common material: earlier work either stops at effective symmetrised tensors [12], or reproduces the exact polar tensor only in principle – with a special lattice, exact only for a circular cloak in the infinite-cell limit [10]. We report both the gains of the microstructured cloak and the fraction of the ideal performance that survives homogenisation.

The paper is organised as follows. Section 2 sets up the half-space problem, the BGM push-forward for the triangular carpet, the cell discretisation, and the JAX-FEM solver and metrics. Section 3 introduces the neural-field design pipeline. Section 4 reports single-frequency results against the ideal and symmetrised baselines. Section 5 extends the design to a broadband objective. Section 6 constrains the cloak to realisable microstructures and validates the single-material realisation. Section 7 discusses the results and their limitations, and Section 8 concludes.

2. Problem setup and FEM baseline

Figure 1 summarises the computational configuration. A time-harmonic vertical point force applied on the traction-free surface of an isotropic half-space (λ, μ, ρ) launches a Rayleigh wave that propagates horizontally towards a triangular surface notch. The rectangular domain of width $W = 12.5\, b$ and depth H is truncated by Perfectly Matched Layer (PML) absorbing regions on the two lateral sides and the bottom; the top surface remains traction-free. The inset (Fig. 1b) shows the cloak geometry in detail: the triangular defect of depth a and half-width c is enclosed by the outer cloak boundary at depth b, whose two inclined faces define the inner and outer triangle boundaries z_1 and z_2; the two mirror phases of the cloak (related by the sign change of the shear F_{21} across the centreline $X_1 = 0$) are shaded separately.

2.1. Governing equations

We consider a homogeneous isotropic half-space with material properties (λ, μ, ρ), where λ and μ are the Lamé coefficients and ρ the mass density, and spatial coordinates $\mathbf{X} = (X_1, X_2)$ for the reference domain, with the free surface on $X_2 = 0$. For in-plane surface waves, i.e. Rayleigh waves, propagating along the horizontal X_1 direction, the governing Navier elastodynamic equation the following form

$$\nabla_X \cdot \big(\mathbf{C} : \nabla_X \mathbf{U}\big) = \rho\, \mathbf{U}_{tt}, \tag{1}$$

where $\mathbf{C}$ is the isotropic fourth-order elasticity tensor, $\mathbf{U} = (U_1, U_2)$ is the displacement, and $\mathbf{U}_{tt}$ denotes its second time derivative. Under the assumption of plane-strain elasticity, the elastic tensor can be written in Voigt notation $\{1, 2, 6\} = \{11, 22, 12\}$ as

$$C_{IJ} = \begin{bmatrix} \lambda+2\mu & \lambda & 0 \\ \lambda & \lambda+2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}, \qquad I, J = 1, 2, 6. \tag{2}$$

We apply a pointwise invertible transformation Ξ that maps the reference configuration (virtual domain) $\mathbf{X} \in \Psi$ to the deformed region (physical domain) $\mathbf{x} = \Xi(\mathbf{X}) \in \psi$ and the remaining domain to itself. Given $\mathbf{x} = (x_1, x_2)$ the coordinates of the physical domain, the transformation gradient and its determinant are

$$\mathbf{F} = \nabla_X \mathbf{x} = \begin{pmatrix} \dfrac{\partial x_1}{\partial X_1} & \dfrac{\partial x_1}{\partial X_2} \\ \dfrac{\partial x_2}{\partial X_1} & \dfrac{\partial x_2}{\partial X_2} \end{pmatrix}, \qquad J = \det(\mathbf{F}). \tag{3}$$

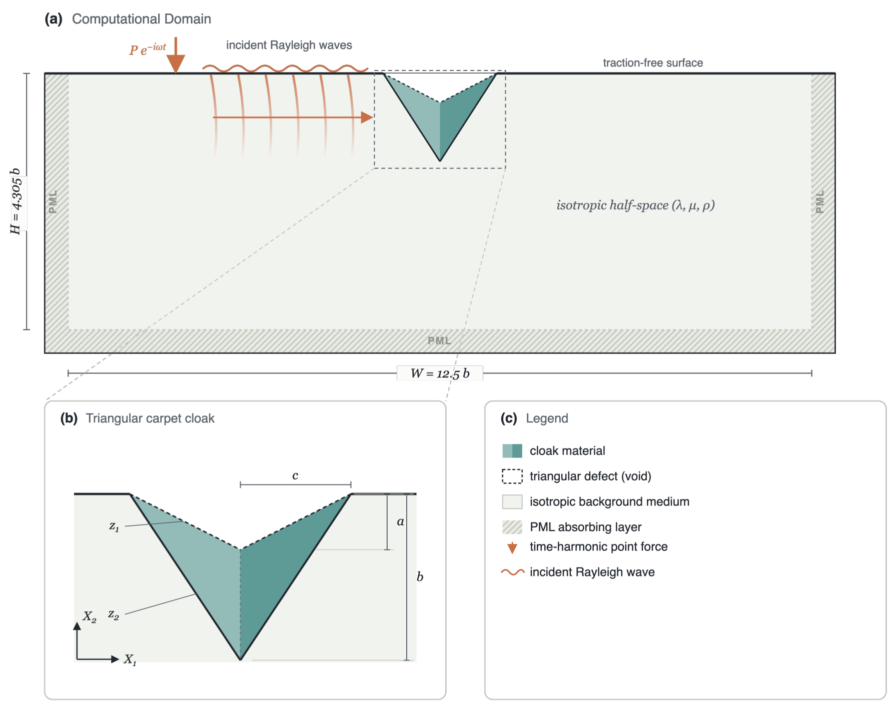


Figure 1: Problem configuration. **(a)** Computational domain: an isotropic elastic half-space (λ, μ, ρ) of width $W = 12.5\,b$ is truncated by PML absorbing layers on the lateral and bottom boundaries; the top surface is traction-free. A time-harmonic vertical point force (red arrow) applied upstream generates Rayleigh waves (orange wavefronts) that propagate towards the triangular cloak (teal). **(b)** Cloak geometry: the surface notch (void) has depth a and half-width c; the outer cloak boundary sits at depth $b = 3a$. Dashed lines mark the inner and outer triangle boundaries $z_1(X_1)$ and $z_2(X_1)$; the two mirror phases of the cloak are shaded in teal.

Equation (1) is not form-invariant under an arbitrary transformation Ξ [8]: the transformed material class depends on the gauge $\mathbf{U}(\Xi(\mathbf{X})) = \mathbf{A}\,\mathbf{u}(\mathbf{x})$ linking the virtual and physical displacements, with $\mathbf{A}$ a non-singular matrix. The choice $\mathbf{A} = \mathbf{F}$ leads to the Willis setting, which guarantees a symmetric stress tensor but requires two additional third-order coupling tensors and a tensorial mass density [8, 2, 21], physically replicable only within narrow frequency bands by resonant microstructures. For this reason we adopt the Cosserat setting $\mathbf{A} = \mathbf{I}$, following Norris and Shuvalov [2] and Brun et al. [1], for which the governing equation in the physical domain retains the Navier form

$$\nabla_x \cdot \left(\mathbf{c}^{\text{eff}} : \nabla_x \mathbf{u}\right) = \rho^{\text{eff}}\,\mathbf{u}_{tt}, \tag{4}$$

with the transformed mechanical parameters given by the Brun–Guenneau–Movchan (BGM) transformation relations

$$c^{\text{eff}}_{ijkl} = J^{-1}\,C_{IjKl}\,F_{iI}\,F_{kK}, \qquad \rho^{\text{eff}} = \rho\,J^{-1}. \tag{5}$$

The transformed tensor preserves the major symmetries ($c^{\text{eff}}_{ijkl} = c^{\text{eff}}_{klij}$) but not the minor ones,

$$c^{\text{eff}}_{ijkl} \neq c^{\text{eff}}_{jikl} \neq c^{\text{eff}}_{ijlk} \neq c^{\text{eff}}_{jilk}, \tag{6}$$

except for special cases such as conformal transformations. The medium is nonetheless described by a single fourth-order, non-symmetric and, eventually, inhomogeneous elastic tensor with scalar density: the stress $\sigma_{ij} = c^{\text{eff}}_{ijkl}\,\partial u_k/\partial x_l$ acts on the *full* displacement gradient – not the symmetric strain – and is itself non-symmetric (polar medium). All computations below are carried out in the frequency domain with the time-harmonic convention $\mathbf{u}(\mathbf{x}, t) = \mathrm{Re}\big[\mathbf{u}(\mathbf{x})\,e^{-\mathrm{i}\omega t}\big]$, which replaces $\mathbf{u}_{tt}$ by $-\omega^2\mathbf{u}$.

2.2. Transformation elasticity for the triangular carpet cloak

The cloak follows the pinched-carpet construction of Chatzopoulos et al. [12]. With the free surface on $X_2 = 0$ and the cloak centred at $X_1 = 0$, the inner (defect) and outer cloak boundaries are the triangles

$$z_1(X_1) = \frac{a}{c}\,|X_1| - a, \qquad z_2(X_1) = \frac{b}{c}\,|X_1| - b, \tag{7}$$

where a is the defect depth, b the cloak depth, and c the surface half-width. The shear map

$$x_1 = X_1, \qquad x_2 = \frac{z_2(X_1) - z_1(X_1)}{z_2(X_1)}\,X_2 + z_1(X_1) \tag{8}$$

compresses the triangular region above z_2 onto the cloak region between z_1 and z_2, opening the notch. Its gradient is piecewise constant,

$$\mathbf{F} = \begin{bmatrix} 1 & 0 \\ F_{21} & F_{22} \end{bmatrix}, \qquad F_{21} = \operatorname{sign}(X_1)\,\frac{a}{c}, \qquad F_{22} = \frac{b-a}{b} = J. \tag{9}$$

For an isotropic background (λ, μ) the BGM push-forward (5) then yields, in augmented Voigt ordering $[\sigma_{11}, \sigma_{22}, \sigma_{12}, \sigma_{21}]$, the effective stiffness

$$\mathbf{c}_{\text{eff}} = \begin{bmatrix} \dfrac{2\mu{+}\lambda}{F_{22}} & \lambda & 0 & \dfrac{F_{21}}{F_{22}}(2\mu{+}\lambda) \\ \lambda & \dfrac{F_{21}^2\mu + F_{22}^2(2\mu{+}\lambda)}{F_{22}} & \dfrac{F_{21}}{F_{22}}\,\mu & F_{21}(\lambda{+}\mu) \\ 0 & \dfrac{F_{21}}{F_{22}}\,\mu & \dfrac{\mu}{F_{22}} & \mu \\ \dfrac{F_{21}}{F_{22}}(2\mu{+}\lambda) & F_{21}(\lambda{+}\mu) & \mu & \dfrac{F_{21}^2(2\mu{+}\lambda) + F_{22}^2\mu}{F_{22}} \end{bmatrix}, \tag{10}$$

together with the constant effective density

$$\rho^{\text{eff}} = \frac{\rho}{J} = \frac{b}{b-a}\,\rho. \tag{11}$$

Unlike radial blow-up cloaks, whose tensors are singular at the inner boundary, the triangular tensor is non-singular and uniform within each half of the cloak (the shear F_{21} changes sign across the centreline $X_1 = 0$), so the ideal cloak consists of just two mirror-image material phases. The price is that *three* minor symmetry pairs are broken independently ($c_{11,12} \neq c_{11,21}$, $c_{22,12} \neq c_{22,21}$, $c_{12,12} \neq c_{21,21}$), and these are the entries that symmetrisation strategies average away by hand.

2.3. Realisable Cauchy parameterisation

Throughout this paper the designed cloak is parameterised by four free orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ (4CM) and a scalar density, obtained by direct numerical search through the differentiable FEM. This class is dictated by the microstructures that must ultimately realise the cloak (Sec. 6): square two-phase unit cells invariant under the point group D_2, whose two edge-aligned mirror axes fix the form of the homogenised stiffness. Writing the plane-strain law in Voigt notation $\{1, 2, 6\} = \{11, 22, 12\}$, a general symmetric Cauchy tensor has six independent constants,

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}C_{11} & C_{12} & C_{16}\\ C_{12} & C_{22} & C_{26}\\ C_{16} & C_{26} & C_{66}\end{bmatrix} \begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ 2\varepsilon_{12}\end{bmatrix}. \tag{12}$$

The edge mirror $(x_1, x_2) \mapsto (x_1, -x_2)$ reverses the shear strain, so requiring invariance $\mathbf{C} = \mathbf{TCT}$ with $\mathbf{T} = \mathrm{diag}(1, 1, -1)$ forces $C_{16} = C_{26} = 0$ and reduces the stiffness to the block-diagonal orthotropic form

$$\mathbf{C}_{\mathrm{4CM}} = \begin{bmatrix}C_{11} & C_{12} & 0\\ C_{12} & C_{22} & 0\\ 0 & 0 & C_{66}\end{bmatrix}, \tag{13}$$

whose four moduli are exactly the per-cell design variables.

2.4. Cell discretisation

We discretise the cloak region into a regular Cartesian grid of $n_x \times n_y$ square cells (Fig. 2). Cells whose centre lies between the inner and outer triangles (7) are assigned a piecewise-constant effective stiffness and the density (11). Because the triangular transformation is affine in Cartesian coordinates, the tensor (10) is passed to the FEM solver without any rotation to a local frame. The ideal cloak tensor (10) is the full 4×4 augmented-Voigt polar tensor with ten independent components. Because real materials are Cauchy (minor-symmetric), the per-cell design variable is four free orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and a scalar density, which are searched numerically by the differentiable optimiser. A systematic investigation of grid resolution established 14×10 as the optimal discretisation, balancing approximation fidelity against per-iteration computational cost; this resolution is used in all single-frequency experiments reported below.

2.5. Simulation domain and FEM solver

We consider a rectangular computational domain of width $W = 12.5\,b$ and depth $H = 4.305\,b$, truncating the half-space. The top boundary is traction-free; PML absorbing layers terminate the two lateral boundaries and the bottom. A triangular surface notch of depth $a = 0.0774\,H$ and half-width $c = 0.1545\,H$ is centred on the free surface, coated by a cloak of depth $b = 3a$. The background medium is isotropic with $\rho = 1600\,\mathrm{kg/m^3}$, shear speed $c_s = 300\,\mathrm{m/s}$, and $c_p = \sqrt{3}\,c_s$ ($\nu = 1/4$, $\lambda = \mu = 144\,\mathrm{MPa}$,

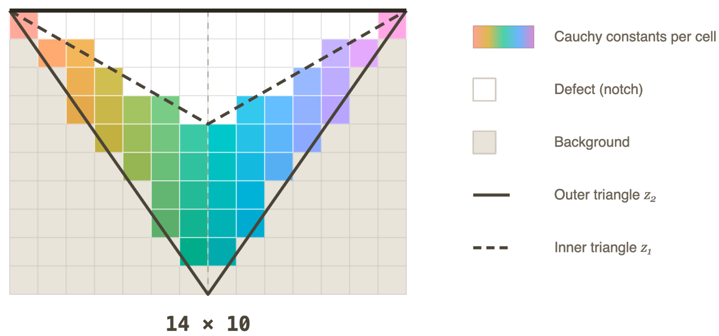


Figure 2: Cartesian cell decomposition of the triangular cloak at the 14×10 resolution used in all single-frequency experiments. Coloured cells carry per-cell Cauchy constants $(C_{11}, C_{12}, C_{22}, C_{66}, \rho)$; white cells lie inside the defect (void notch); grey cells are background. The solid line is the outer cloak boundary z_2 and the dashed line is the inner boundary z_1.

plane strain). The excitation is a time-harmonic vertical point force applied on the free surface upstream of the cloak, generating Rayleigh waves that propagate along the surface towards the defect. Frequencies are reported in the dimensionless form $f^\star = f\,b/c_R$, where $c_R \approx 0.92\,c_s$ is the Rayleigh wave speed of the background: at $f^\star = 1$ the Rayleigh wavelength equals the cloak depth b. The nominal design frequency is $f^\star = 2$.

The frequency-domain elastodynamic equations are solved with complex-valued degrees of freedom and Rayleigh-damping PML through a JAX-FEM finite-element solver [13] on a triangular mesh with distance-based refinement near the cloak boundaries ($\sim$246,000 elements, $\sim$124,000 nodes). The entire pipeline from cell material parameters through assembly and sparse solve to the cloaking loss is differentiable via implicit adjoint differentiation through the linear FEM solve, enabling gradient-based optimisation of the per-cell stiffness and density.

2.6. Cloaking metrics

We use three relative L^2 metrics. The *right-boundary distortion*

$$\mathcal{D}_\mathrm{r} = 100\,\frac{\|\mathbf{u}_\mathrm{cloak} - \mathbf{u}_\mathrm{ref}\|_{L^2(\partial\Omega_\mathrm{right})}}{\|\mathbf{u}_\mathrm{ref}\|_{L^2(\partial\Omega_\mathrm{right})}}\,, \tag{14}$$

measures the transmission-side mismatch with respect to the defect-free reference. The *outside-cloak distortion* $\mathcal{D}_\mathrm{out}$ replaces the boundary integral by an integral over $\Omega_{\mathrm{phys}} \setminus \overline{\Omega}_{\mathrm{cloak}}$ and captures the global wavefield perturbation; it is closely related to the objective used by Wang et al. [11]. In broadband sections we also report the dimensionless cloak ratio $\langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\mathrm{ref}}|\rangle$ over the free surface beyond the cloak footprint, following Chatzopoulos et al. [12]; this is the cloaking objective used in the microstructure-constrained pipeline.

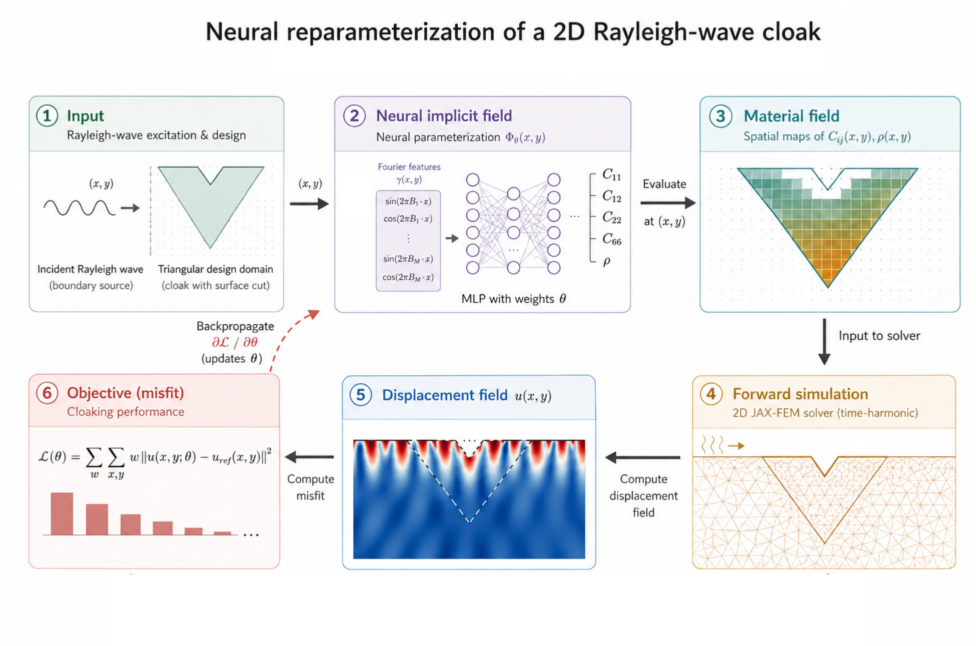


Figure 3: Neural-field design pipeline. Cell-centre coordinates are mapped through a Fourier positional encoding into a four-layer MLP that predicts the four free Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and the density. The piecewise-constant material assignment is expanded to the FEM quadrature points and the cloaking objective is differentiated back through the implicit JAX-FEM adjoint into the network weights.

3. Neural-field design pipeline

3.1. Coordinate-conditioned material network

We reparameterise the cloak material field through a coordinate-based multilayer perceptron (MLP)

$$\Phi_\theta \colon (x, y) \;\mapsto\; (\mathbf{C}_{4\mathrm{CM}}(x, y),\, \rho(x, y)), \tag{15}$$

where θ denotes the network weights, $\mathbf{C}_{4\mathrm{CM}} \in \mathbb{R}^4$ contains the four free Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ — the design variables directly searched by the optimiser — and ρ is a positive scalar density. The network is evaluated at each cloak-cell centre to produce the piecewise-constant material assignment that is then expanded to the FEM quadrature points. A four-layer MLP with 256 hidden units ($\sim$200,000 weights) is used in all experiments; we adopt tanh activations and the Fourier positional encoding of Tancik et al. [16] so that the network can represent sharp material transitions near the cloak boundaries.

Figure 3 shows the architecture and the data flow through the differentiable FEM. Gradients $\partial\mathcal{L}/\partial\theta$ are obtained by backpropagating through the JAX-FEM adjoint solve and into the network.

3.2. Transformation-elasticity prior

The network does not learn the cloak from scratch. The four Cauchy moduli per cell are initialised at a starting point $\mathbf{C}^{(0)}_{4\mathrm{CM}}$ and then freely optimised. We tried two choices for this starting point – the arithmetic-mean symmetrisation of the ideal tensor (10) and the mean of the microstructure dataset (Sec. 6) – and found no statistically significant difference in the converged results, indicating that the design is driven by the cloaking objective rather than by the initialisation. Concretely, the last layer is scaled near zero ($\mathbf{W} \leftarrow 0.01\,\mathbf{W},\ \mathbf{b} \leftarrow \mathbf{0}$) and the network output is applied as a *multiplicative residual* with default output scale $\epsilon = 0.1$:

$$\mathbf{C}_{4\mathrm{CM}}(x,y) = \mathbf{C}^{(0)}_{4\mathrm{CM}}(x,y) \odot \left(1 + \epsilon\,\Phi^{(C)}_\theta(x,y)\right), \tag{16a}$$

$$\rho(x,y) = \rho^{(0)}(x,y)\left(1 + \epsilon\,\Phi^{(\rho)}_\theta(x,y)\right). \tag{16b}$$

The multiplicative form makes the relative perturbation spatially uniform regardless of the orders of magnitude separating the stiffness and density entries within each cloak phase. An additive variant is also implemented but produces more poorly conditioned gradients in practice.

3.3. Physical admissibility of the material output

The multiplicative residual (16) places no bound on the sign or the anisotropy of the moduli: an unconstrained MLP can drive a stiffness negative, break the positive-definiteness of the Cauchy tensor, or reach an anisotropy ratio no real microstructure can realise. Rather than police these violations with penalty terms after the fact, we build them into the decoding, so that *every* network output — at initialisation, along the optimisation path, and at convergence — is a physically admissible material by construction. The optimiser therefore searches only over the manifold of valid tensors and never has to be projected back onto it.

The network still emits four unbounded raw channels $\mathbf{r} = (r_{11}, r_{22}, r_{66}, r_{12}) = \Phi_\theta^{(C)}(x, y)$ per cloak cell, but these are mapped to the moduli through three guarantees.

(i) Positive diagonal and shear. The three positive-definite entries are set by a log-space (hence strictly positive) multiplicative residual,

$$C_{ii} = C_{ii}^{(0)} \exp(\epsilon\, r_{ii}), \qquad ii \in \{11, 22, 66\}, \tag{17}$$

which keeps them positive for any real r_{ii} and recovers $C_{ii}^{(0)}$ at $\mathbf{r} = \mathbf{0}$.

(ii) Bounded anisotropy. To exclude ratios beyond what the microstructure can attain, the diagonal ratio C_{11}/C_{22} is confined to $[1/R,\, R]$ while its geometric mean $\sqrt{C_{11}C_{22}}$ is held fixed. Writing $g = \frac{1}{2}\log(C_{11}/C_{22})$, a tanh squash on the half-log ratio,

$$g_b = \tfrac{1}{2}\log R\, \tanh\!\Big(\frac{g}{\frac{1}{2}\log R}\Big), \qquad C_{11}, C_{22} = \sqrt{C_{11}C_{22}}\, e^{\pm g_b}, \tag{18}$$

bounds the ratio smoothly to $[1/R, R]$ with R the anisotropy cap (we use $R = 30$, chosen above the 99th percentile of the microstructure dataset with margin).

(iii) Positive-definite coupling. The off-diagonal C_{12} is written through a correlation coefficient $\rho_c \in (-\kappa, \kappa)$ rather than directly,

$$C_{12} = \rho_c \sqrt{C_{11}C_{22}}, \qquad \rho_c = \kappa\, \tanh\!\big(r_{12} + \phi_0\big), \tag{19}$$

with $\kappa < 1$ (we use $\kappa = 0.99$). Because $|\rho_c| < \kappa < 1$, the orthotropic block stays strictly positive-definite, $\det = C_{11}C_{22}(1 - \rho_c^2) > 0$, for any r_{12}, and C_{12} is free to change sign. The offset $\phi_0 = \operatorname{arctanh}(\rho_c^{(0)}/\kappa)$ centres the map so that $\mathbf{r} = \mathbf{0}$ again recovers the initial coupling $C_{12}^{(0)}$. Non-cloak cells bypass the decoding and keep their background values exactly.

Together, (17)–(19) turn the raw network outputs into a smooth, everywhere-differentiable surjection onto the set of positive-definite, bounded-anisotropy Cauchy tensors, so the transformation-elasticity initialisation is recovered at $\mathbf{r} = \mathbf{0}$ and every gradient step lands on an admissible design.

3.4. Loss function

The optimisation minimises a *magnitude-band integral*: the mean-squared relative error of the displacement magnitude over a strip of depth $d = 0.5\,b$ (half the cloak depth) below the free surface, downstream of and excluding the cloak/defect footprint,

$$\mathcal{L}_{\text{cloak}}(\theta) = \frac{1}{|\mathcal{B}|}\int_{\mathcal{B}}\left(\frac{|\mathbf{u}_{\text{cloak}}|}{|\mathbf{u}_{\text{ref}}|} - 1\right)^2 \mathrm{d}A, \tag{20a}$$

$$\mathcal{B} = \left\{(x,y) \in \Omega_{\text{phys}} \setminus \overline{\Omega}_{\text{cloak}} : y_{\text{top}} - d \le y \le y_{\text{top}},\; x > x_c\right\}. \tag{20b}$$

where $\mathbf{u}_{\text{cloak}}$ and $\mathbf{u}_{\text{ref}}$ are the cloaked and defect-free reference fields and x_c is the downstream edge of the cloak. Comparing magnitudes rather than the complex fields themselves makes the objective phase-invariant, which is a deliberate advantage here: a wave that is diverted around the defect follows a slightly longer path and arrives with a small phase lag relative to the flat-surface reference. An explicit neighbour-smoothness term used in raw-parameter baselines [11] is unnecessary here: the neural field is spatially continuous by construction, and its smoothness is controlled directly through the Fourier positional encoding [16]. The number and bandwidth of the Fourier features set the spatial frequency content the network can represent—low frequencies yield smooth, gently varying material fields, whereas adding higher-frequency components lets the field resolve sharper transitions near the cloak boundaries.

4. Single-frequency results

All single-frequency results are reported at the design frequency $f^\star = 2.0$ through the dimensionless *cloak ratio*

$$\eta \;=\; \frac{\langle|\mathbf{u}|\rangle}{\langle|\mathbf{u}_{\text{ref}}|\rangle} \tag{21}$$

of the mean surface-displacement magnitude on the free surface beyond the cloak footprint to that of the defect-free reference (Sec. 2), so that $\eta \to 1$ is perfect cloaking and $|1-\eta|$ measures the residual distortion. We build up the design in three steps of increasing material freedom: a single homogeneous realisable material (Sec. 4.1), a handful of independent materials (Sec. 4.2), and finally a cell grid selected by a convergence sweep (Secs. 4.3–4.4).

4.1. A single realisable material already beats symmetrisation

We first ask how far a *single* realisable material can go. Both cloak phases are collapsed onto one orthotropic 4CM tile (a 1×1 cell filling the whole cloak) and its four Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and density are optimised through the differentiable FEM at $f^\star = 2.0$. The natural point of comparison is the closed-form arithmetic-mean symmetrisation of Chatzopoulos et al. [12]: the ideal polar tensor (10) made minor-symmetric by averaging each broken pair, instantiated on the same mesh and solved with the same FEM. Both are single homogeneous Cauchy materials, so this is a like-for-like test of hand symmetrisation against gradient-based optimisation.

Table 1 reports the outcome. The symmetrised tensor recovers only $\eta = 0.63$ and actually distorts the near field more than the bare notch (the ideal-material minor-symmetry violation is large at $f^\star = 2$). A single optimised orthotropic material instead reaches $\eta = 0.85$ — a substantial jump towards the ideal continuous polar cloak ($\eta = 0.999$) obtained from just one degree of material freedom per phase, and clear evidence that optimising inside the Cauchy class outperforms symmetrisation. It is, however, still far from perfect: one material cannot reproduce the depth variation of the ideal tensor. Figure 4 confirms that the advantage is concentrated around the design frequency, as expected for a single-frequency optimisation.

Table 1: Single-frequency cloak ratio η at $f^\star = 2.0$ for a single homogeneous realisable material. The optimised orthotropic tile outperforms the closed-form symmetrisation of Chatzopoulos et al. [12], yet neither attains the ideal continuous polar cloak.

Configuration	η	$\lvert 1-\eta \rvert$
Uncoated notch (no cloak)	0.37	0.63
Symmetrised ideal (Chatzopoulos, closed-form Cauchy)	0.634	0.366
Single optimised material (1×1, 4CM)	0.850	0.150
Ideal continuous $\mathbf{c}_{\text{eff}}$ (polar, unrealisable)	0.999	0.001

4.2. How many distinct materials does the cloak need?

Giving each phase more freedom closes most of the remaining gap. Splitting the cloak into a coarse $n_x \times n_y$ grid of independent orthotropic tiles and optimising all of them jointly, we find that even a 2×2 decomposition — just *four* realisable materials — lifts the cloak ratio to $\eta = 0.962$, i.e. within

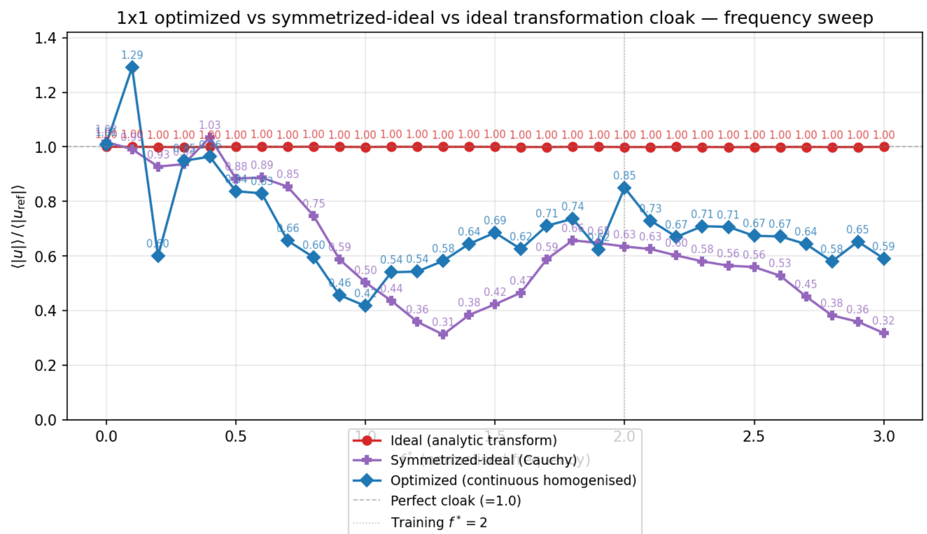


Figure 4: Frequency sweep of the cloak ratio $\eta = \langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\text{ref}}|\rangle$ for the single-material designs: the ideal continuous transformation cloak (red, $\eta \approx 1$ at every frequency), the closed-form symmetrised tensor of Chatzopoulos et al. [12] (purple), and the single optimised 1×1 material (blue). At the training frequency $f^\star = 2$ the optimised material recovers $\eta = 0.85$ against 0.63 for symmetrisation.

4% of a perfect cloak (Table 2). The improvement is markedly anisotropic: subdividing in depth helps far more than in the propagation direction (1×2 reaches $\eta = 0.92$ whereas 2×1 actually drops to $\eta = 0.78$), reflecting that the ideal cloak material varies mainly with depth through the shear map (8). Beyond four materials the gain per added cell is small, motivating a more systematic grid study.

Table 2: Cloak ratio η at $f^{\star} = 2.0$ for coarse decompositions, each cell an independent orthotropic material optimised jointly. Four materials (2×2) already reach $\eta = 0.962$.

Decomposition $n_x \times n_y$	Materials	η	$\lvert 1-\eta \rvert$
1×1	1	0.857	0.143
2×1	2	0.780	0.220
1×2	2	0.920	0.080
3×1	3	0.939	0.061
2×2	**4**	**0.962**	**0.038**
3×2	6	0.964	0.036

4.3. Grid selection by a cell-convergence sweep

To push towards perfect cloaking we refine the grid, but the resolution must be chosen so that cloaking has *converged* while the per-iteration cost stays low. We therefore run a single-frequency cell sweep over seven Cartesian grids from 5×4 to 32×24, optimising each to convergence and recording the cloak ratio (Table 3).

A key point is that materials are assigned only to cells whose centre falls between the inner and outer triangles (7), so an $n_x \times n_y$ grid uses far fewer than $n_x n_y$ materials: the triangular cloak occupies only a fraction of its Cartesian bounding box. The 14×10 grid nominally spans 140 cells but places only **46 cells inside the cloak region**; these 46 are the actual design variables, the remaining 94 being background (see the "cloak cells" column of Table 3).

The cloak ratio saturates on a plateau at $\eta \approx 0.99$ ($\lvert 1-\eta \rvert \approx 0.01$) by 14×10: the three finer grids (20×15, 26×20, 32×24) give no further improvement — and even a marginal regression from FEM cost and optimisation noise — while their cell counts, and hence solve cost, grow several-fold. The 14×10 grid is the smallest decomposition sitting firmly on this converged sub-1%

plateau, and we adopt it for all subsequent single-frequency work: it is stable (on the plateau), cheap (only 46 active materials), and fast to optimise.

Table 3: Cell-convergence sweep at $f^\star = 2.0$: cloak ratio η versus grid resolution. "Cloak cells" counts the cells inside the triangular cloak (the active design variables, far fewer than $n_x n_y$). Cloaking plateaus at $\eta \approx 0.99$ from 14×10 onward.

Grid $n_x \times n_y$	Total cells	Cloak cells	η	$\lvert 1-\eta \rvert$
5×4	20	7	0.944	0.056
8×6	48	16	0.943	0.057
10×8	80	28	0.977	0.023
14×10	**140**	**46**	**0.990**	**0.010**
20×15	300	100	0.990	0.010
26×20	520	176	0.988	0.012
32×24	768	256	0.988	0.012

4.4. Near-perfect cloaking on the 14×10 *grid*

The adopted 14×10 design (46 optimised in-cloak materials) reaches $\eta = 0.990$, effectively closing the gap to the ideal continuous cloak. Figure 5 compares the real-displacement magnitude of the optimised cloak against three references on the same mesh: the defect-free reference, the uncloaked notch, and the ideal continuous polar $\mathbf{c}_{\text{eff}}$. The uncloaked notch back-scatters the incident Rayleigh wave and casts a clear shadow downstream; the optimised 14×10 cloak restores the surface wavefield to the reference, matching the ideal continuous cloak panel while using only realisable orthotropic Cauchy materials. The only visible residual is a slight phase shift of the optimised solution: Figure 6 zooms into the cloak region and shows that, whereas the ideal continuous cloak keeps its near-surface wavefronts phase-aligned with the reference, the optimised 14×10 cloak carries them $\approx 0.18\lambda$ ($\approx 65^\circ$) later. This lag is a direct and benign consequence of the phase-invariant magnitude loss (20): the objective constrains only the amplitude envelope, leaving a constant phase offset — the small extra path of a wave routed around the notch — unpenalised. Figure 7 shows the corresponding optimised per-cell moduli $(C_{11}, C_{22}, C_{12}, C_{66})$ and density over the 46 cloak cells.

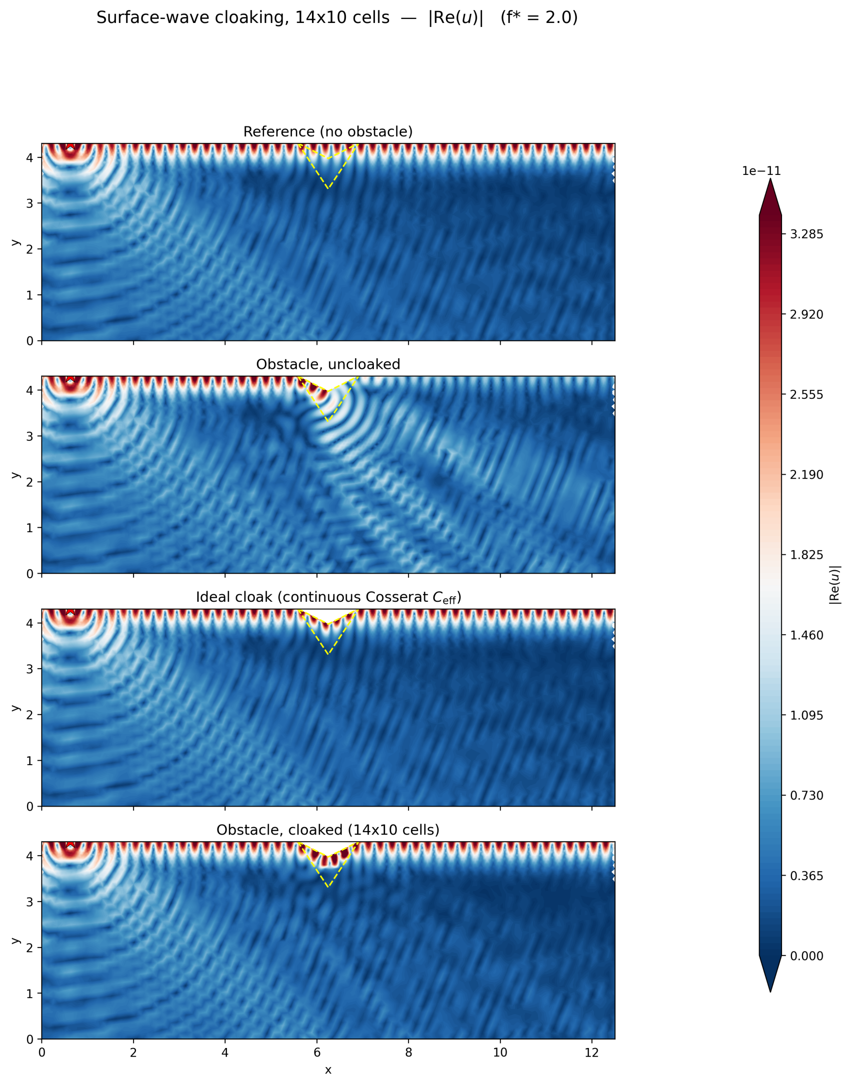


Figure 5: Real-displacement magnitude $|\mathrm{Re}\,\mathbf{u}|$ at $f^\star = 2.0$ (top to bottom): defect-free reference, uncloaked notch, ideal continuous polar cloak $\mathbf{c}_{\mathrm{eff}}$, and the optimised 14×10-cell cloak (46 in-cloak materials, $\eta = 0.990$). The optimised cloak reconstructs the reference wavefield on the transmission side and is visually indistinguishable from the ideal continuous cloak, with the downstream shadow of the uncloaked notch removed. Dashed lines mark the cloak/defect boundary.

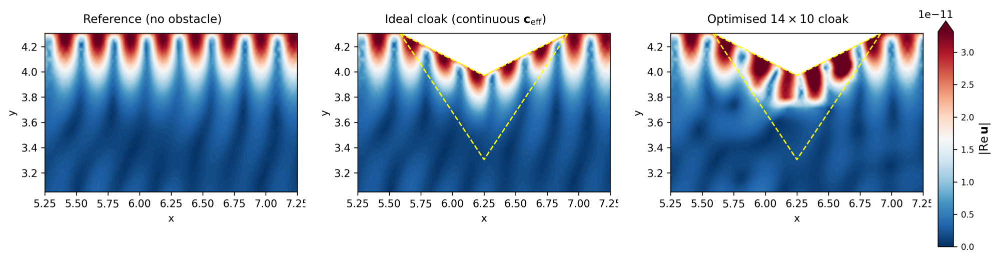


Figure 6: Zoom on the cloak region ($|\mathrm{Re}\,\mathbf{u}|$ at $f^\star = 2.0$, shared colour scale): defect-free reference (left), ideal continuous polar cloak (centre), and optimised 14×10 cloak (right); dashed lines outline the inner (defect) and outer cloak boundaries. All three reproduce the same amplitude envelope, but the optimised cloak leaves its near-surface wavefronts slightly phase-shifted ($\approx 0.18\lambda$, $\approx 65°$) relative to the reference and the ideal cloak — a constant phase offset left unconstrained by the phase-invariant magnitude loss, not a cloaking error.

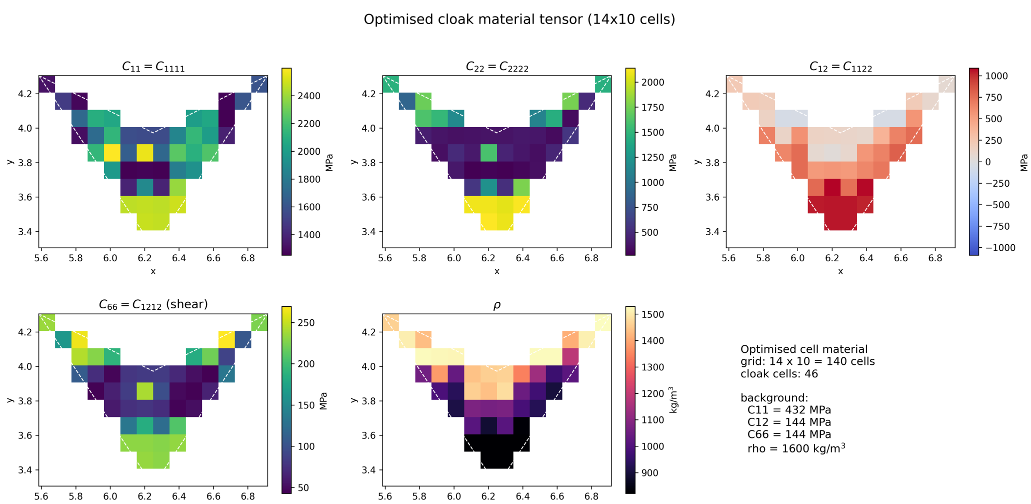


Figure 7: Optimised per-cell material of the 14×10 single-frequency cloak: the four orthotropic Cauchy moduli $C_{11}, C_{22}, C_{12}, C_{66}$ and the density ρ over the 46 cells inside the cloak region (background cells masked). Only these 46 materials are optimised.

5. Broadband multifrequency optimisation

The single-frequency designs of Sec. 4 are optimised at one frequency and, being tuned to it, cloak well only in a narrow neighbourhood of $f^\star = 2$: away from the design frequency the cloak ratio falls off markedly (Fig. 4). Because the ideal triangular tensor (10) is frequency-independent and cloaks exactly at every frequency, this narrowness is a limitation of the single-frequency *objective*, not of the material class. We therefore train a single design against a band of frequencies at once.

5.1. Band objective

The broadband loss is the average of the single-frequency magnitude-band integral (20) over a finite set $\mathcal{S}$ of frequencies sampling the target band,

$$\mathcal{L}_{\text{band}}(\theta) = \frac{1}{\sum_{f\in\mathcal{S}} w_f} \sum_{f\in\mathcal{S}} w_f \, \mathcal{L}_{\text{cloak}}(\theta; f), \tag{22}$$

where $\mathcal{L}_{\text{cloak}}(\theta; f)$ is (20) evaluated at frequency f against the corresponding defect-free reference field $\mathbf{u}_{\text{ref}}(f)$. Equal weights $w_f = 1$ minimise the mean in-band distortion; the weights are free to emphasise particular frequencies, and a worst-case (min–max) variant that up-weights the currently least-cloaked frequency can be used to flatten the response across the band. A single neural field Φ_θ produces the shared per-cell material, so the same design is required to cloak at every $f \in \mathcal{S}$ simultaneously.

The objective (22) places no restriction on which band is targeted: any $\mathcal{S}$ can be chosen, and the optimiser returns the best single realisable design over that band. The only cost is computational — every optimisation step evaluates a forward and an adjoint FEM solve per frequency, so the per-step cost scales linearly with $|\mathcal{S}|$. The frequencies are independent and solved in parallel across a worker pool, bounded in practice only by the memory of the concurrent sparse factorisations.

5.2. Frequency sweep

We optimise the 14×10 design (the same 46 in-cloak Cauchy materials and neural field as in Sec. 4) against the band $f^\star \in [1, 3]$, sampled at seven uniformly spaced frequencies $\mathcal{S} = \{1.0, 1.33, 1.67, 2.0, 2.33, 2.67, 3.0\}$ with equal weights. The macro mesh is sized to resolve the shortest wavelength in the band (at $f^\star = 3$) and reused across all frequencies. To probe generalisation beyond the training frequencies, the converged design is then evaluated on a dense sweep $f^\star \in [0, 4]$ and compared against the uncloaked notch and the ideal continuous polar cloak on the same mesh (Fig. 8).

Across the training band the optimised cloak holds the cloak ratio near unity at every frequency: η stays in the range 0.92–0.99 with an in-band mean of 0.96 and a worst case of $|1 - \eta| \approx 0.08$ at the lower band edge, where the uncloaked notch instead drops to $\eta \approx 0.31$ ($|1 - \eta| \approx 0.69$) near mid-band. The ideal continuous cloak remains at $\eta \approx 1$ throughout, as expected for the frequency-independent triangular tensor. The single design thus recovers most of the single-frequency performance simultaneously over the whole octave-plus band, rather than at one frequency only. Outside the training band the improvement degrades gracefully — $\eta \approx 0.78$ at $f^\star = 3.5$ and ≈ 0.60 at $f^\star = 4.0$ — confirming that the objective (22) broadens the cloaking response over the targeted band rather than trivially across all frequencies.

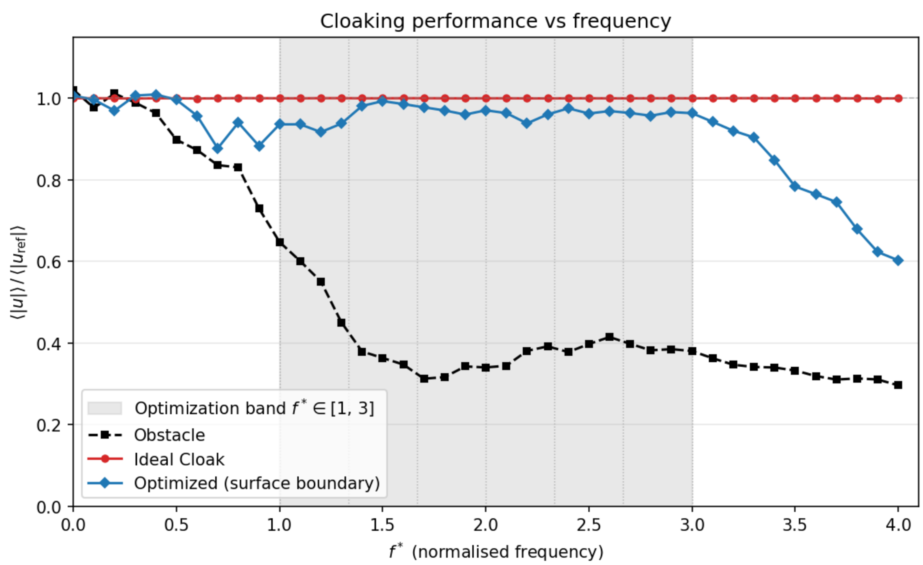


Figure 8: Frequency sweep of the cloak ratio $\eta = \langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\text{ref}}|\rangle$ for the broadband 14×10 design trained on the band $f^\star \in [1, 3]$ (shaded): the ideal continuous transformation cloak (red, $\eta \approx 1$ at every frequency), the uncloaked notch (black, dashed), and the broadband optimised cloak (blue). The optimised design holds $\eta \approx 0.96$ across the entire training band and degrades gracefully outside it.

5.3. Floquet–Bloch dispersion

To verify that the optimised broadband material is physically admissible we compute the Floquet–Bloch dispersion of a vertical strip unit cell cut through the optimised cloak, Bloch-periodic along the surface direction x_1, traction-free at the top and fixed at the bottom, following the construction used for the Rayleigh-wave benchmark [12]. Because the triangular transformation is invariant along x_1, the cell width is unconstrained. Figure 9 reports the lowest surface branches over the reduced Brillouin zone $k_{x_1} \in (0, \pi/L_c]$. The dispersion is real-valued across the target band — the optimised cell supports propagating surface modes and does not rely on locally resonant bandgaps to suppress transmission-side amplitude.

`bloch_dispersion.png`

Figure 9: Floquet–Bloch dispersion of a vertical strip unit cell through the broadband neural-field cloak, Bloch-periodic along the surface direction. Shaded band is the broadband training target.

6. Microstructure-constrained realisation

The cloaks of Sec. 5 are tensor-valued: each cloak cell carries a homogenised $(\mathbf{C}_{4\mathrm{CM}}, \rho)$ that need not correspond to any manufacturable object. We now close the loop to an explicit, single-material realisation, in which every cloak cell is filled with a two-phase cement/void microstructure whose homogenised stiffness reproduces the optimised target, so that the whole cloak can be built from one constituent by varying only the internal pattern from cell to cell. We first constrain the neural-field optimisation to the manifold of realisable microstructures, then compare two generative methods – inverse design and conditional diffusion – for turning each per-cell target into an actual microstructure, and finally exhibit and validate the realised cloak.

6.1. Pipeline

Step 1 – Microstructure dataset.. A catalogue of $N \sim 10^6$ binary 50×50 unit cells is generated by a squared-assembly cellular automaton on a cement/void substrate, following the cellular-automaton generative approach of Nakarmi et al. [20], who assemble a large, topologically diverse library of mesostructures by evolving a pixel grid under neighbour-count birth/death/survival rules.

Step 2 – Homogenisation.. Periodic-FEM homogenisation of each unique cell yields its effective orthotropic stiffness $(C_{11}, C_{22}, C_{12}, C_{66})$ and density ρ_{eff}. Because the squared-assembly cells are invariant under the point group D_2, the homogenised tensor is exactly of the 4CM form (13) that is the design variable throughout the paper, so the descriptor carries the full (orthotropic) stiffness rather than an isotropic projection of it.

Step 3 – Realisability prior.. A Gaussian mixture is fit to the standardised $(\mathbf{C}_{4\mathrm{CM}}, \rho)$ point cloud of the catalogue, and a flat-top threshold τ at the q-th log-density quantile defines the realisable manifold $\mathcal{M}$.

Step 4 – Constrained optimisation.. The neural field of Sec. 3 is trained with the manifold penalty

$$\mathcal{L}_{\mathrm{GMM}} = \sum_{c} \max\big(0,\, \tau - \log p_{\mathrm{GMM}}(\mathbf{C}_{4\mathrm{CM},c}, \rho_c)\big), \tag{23}$$

added to the single-frequency or band cloak loss. The penalty pulls every cell into $\mathcal{M}$ without pinning it to a specific catalogue entry, while the positive-definiteness and bounded-anisotropy guarantees of Sec. 3.3 keep every intermediate design admissible.

Step 5 – Microstructure realisation.. Each optimised per-cell target $(\mathbf{C}_{4\mathrm{CM}}, \rho)$ is turned into an explicit 50×50 cement/void cell by one of the two generative methods of Sec. 6.2 (or a nearest-neighbour dataset baseline).

Step 6 – Evaluation.. The macro FEM is re-solved with every cloak cell carrying the homogenised stiffness of its realised microstructure, and the cloak ratio η (21) is recomputed.

6.2. Two realisation methods

Both methods solve the same per-cell inverse problem – given a target orthotropic stiffness and density, produce a binary 50×50 cell whose homogenisation reproduces it – but approach it from opposite directions.

Inverse design (neural field).. A coordinate network parameterises a continuous density field over the 50×50 cell; it is relaxed during training and driven to a binary pattern through a soft-to-hard schedule, with a connectivity penalty that suppresses disconnected cement islands and floating voids. The cell's periodic-FEM homogenised stiffness is differentiable in the network weights, so the pattern is optimised directly against the per-cell target error. The method is deterministic and returns a single, locally optimal cell for each target.

Diffusion (conditional generative).. A conditional denoising diffusion model is trained on the catalogue to sample binary cells conditioned on a target $(\mathbf{C}_{4\mathrm{CM}}, \rho)$, following the guided-diffusion inverse-design approach of Yang et al. [22]. Because sampling is stochastic, N candidates are drawn per cell and the one whose homogenised stiffness is closest to the target is kept (*best-of-N*). The generative prior keeps every sample inside the manifold of manufacturable cells while exploring a diverse set of realisations for each target.

Figure 10 quantifies how well each method improves its per-cell target across the 100 cloak cells compared to the nearest-neighbour snap. Section 6.3 shows, however, this per-cell stiffness accuracy does not translate directly into whole-cloak performance.

6.3. Results

Table 4 reports the cloak ratio at $f^\star = 2.0$ on the 20×15 macro grid (100 cloak cells) for the three realisation strategies, each evaluated with the homogenised stiffness of the realised cells. The continuous 4CM optimum

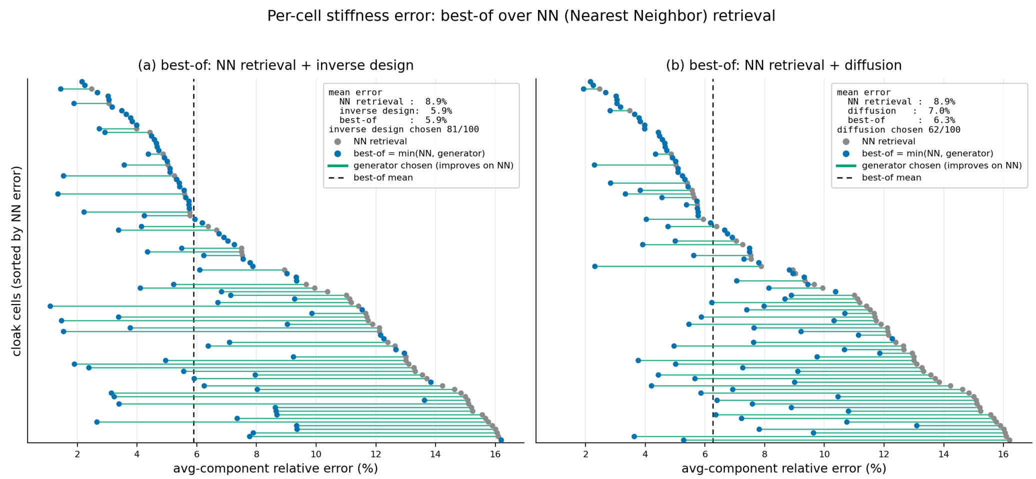


Figure 10: Per-cell stiffness-matching error over the 100 cloak cells for the two realisation methods against NN (Nearest Neighbor) retrieval: (a) NN + inverse design, (b) NN + diffusion. Each row is a cell; the bar links the NN error (grey) to the per-cell best-of selection (blue, the lower-error of the two). Best-of lowers the mean avg-component stiffness error from 8.9% to 5.9% (inverse) and 6.3% (diffusion); cells are sorted by NN error.

reaches $\eta = 0.99$; realising it with actual microstructures costs performance in proportion to how well each method hits its per-cell target. The nearest-neighbour dataset snap retains only $\eta = 0.82$, the neural-field inverse design lifts this to $\eta = 0.94$, and the best-of-N diffusion realisation recovers $\eta = 0.97$ — within 3% of the continuous optimum using only fabricable cement/void cells.

The advantage of the generative realisation is not confined to the design frequency. Figure 11 sweeps $\eta(f)$ across $f^\star \in [1.5, 2.5]$ for the three methods on the matched homogenised cloak: diffusion holds $\eta \gtrsim 0.93$ over $f^\star \in [1.8, 2.2]$ and has the smallest mean in-band distortion ($\langle|1 - \eta|\rangle = 0.13$, against 0.17 for inverse design and 0.25 for nearest neighbour), so better per-cell realisation improves the cloak across the whole band, not only at $f^\star = 2$. We note that a greedy per-cell hybrid — picking, for each cell independently, whichever of the two methods best matches that cell's target — does *not* beat diffusion ($\eta = 0.95$): minimising per-cell stiffness error is not the same as maximising whole-cloak cloaking, because the ratio couples the cells through the shared wave field.

Table 4: Microstructure realisation of the constrained 20×15 cloak at $f^\star = 2.0$ (100 cloak cells): cloak ratio η (homogenised, matched) for the three cell-realisation strategies, against the continuous 4CM optimum and the ideal polar cloak. Best-of-N diffusion recovers $\eta = 0.97$.

Realisation of the 100 cloak cells	η	$\lvert 1-\eta \rvert$
Continuous 4CM optimum (homogenised, GMM-constrained)	0.990	0.010
Nearest-neighbour dataset match	0.816	0.184
Inverse design (neural field)	0.936	0.064
Diffusion (best-of-N)	**0.967**	**0.033**
Ideal continuous polar $\mathbf{c}_{\text{eff}}$ (unrealisable)	0.999	0.001

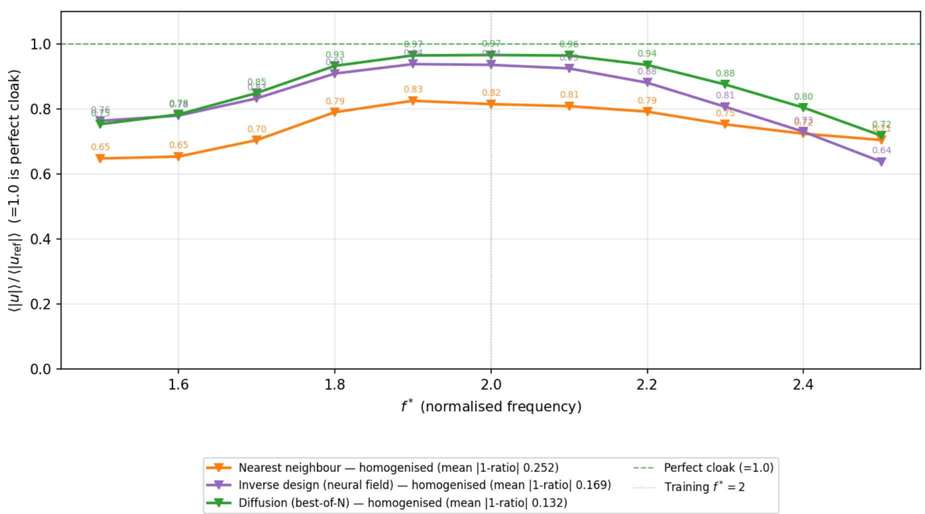
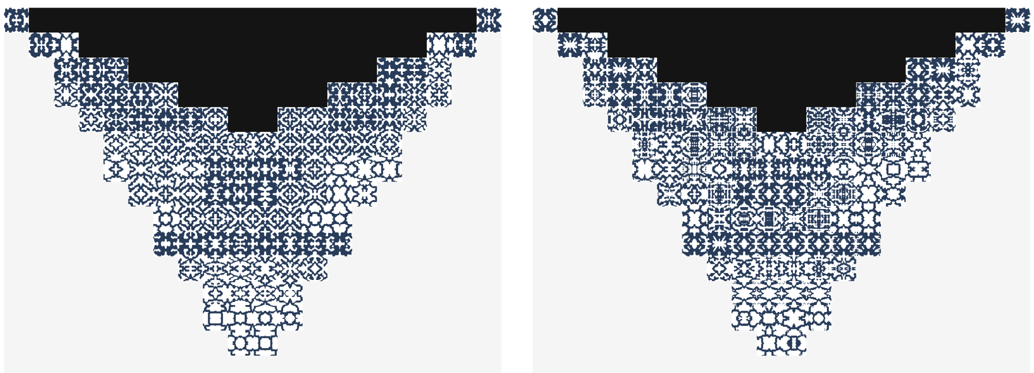


Figure 11: Frequency sweep of the cloak ratio $\eta(f) = \langle|\mathbf{u}|\rangle / \langle|\mathbf{u}_{\text{ref}}|\rangle$ for the three microstructure-realisation methods on the matched homogenised 20×15 cloak: nearest-neighbour dataset snap (orange), neural-field inverse design (purple), and best-of-N diffusion (green). Diffusion stays closest to the perfect-cloak line ($\eta = 1$) across the band and has the smallest mean distortion; the dashed line is the training frequency $f^\star = 2$.

6.4. Realised cloak and validation

Figure 12 shows the explicit cement/void realisation of the cloak: the 100 generated 50×50 cells tiled at the macro-cell positions they occupy inside the triangular carpet, with the surface notch and the surrounding half-space. The whole cloak is a single cement phase perforated by a cell-varying void pattern, densest near the outer boundary and most strongly perforated near the sheared defect tip, mirroring the depth variation of the ideal tensor (10).



(a) Diffusion (best-of-N)

(b) Inverse design (neural field)

Figure 12: Explicit realisation of the triangular carpet cloak from a single cement phase: the 100 generated 50×50 unit cells tiled at the macro-cell positions they occupy in the cloak. Cement is dark, void white; the black inverted triangle is the surface notch and the light grey region is the surrounding half-space. (a) Best-of-N diffusion and (b) neural-field inverse design produce visibly different but equally manufacturable patterns for the same optimised per-cell targets.

On pixel-level validation.. Because the cloak satisfies strong scale separation — the Rayleigh wavelength spans $S = \lambda_R/\delta \approx 190$–$600$ micro-pixels — asymptotic homogenisation guarantees that a mesh-converged, pixel-resolved solve of the full cloak reproduces the homogenised cloak ratio to within $\mathcal{O}\big((d_{\text{cell}}/\lambda_R)^2\big) \approx 2\%$. The matched homogenised η of Table 4 is therefore the physically relevant metric. A brute-force pixel-resolved solve of the whole macro domain is correspondingly redundant, and at the resolution needed to preserve the thin cement ligaments ($\gtrsim 2$ elements per micro-pixel over the entire cloak, $\sim 10^7$–10^8 degrees of freedom) it is computationally infeasible; at attainable resolutions an unstructured mesh aliases the ligaments and the ratio fails to converge — a discretisation artefact rather than a physical loss. We therefore anchor the realisation at the unit-cell level, where each generated cell reproduces its target stiffness under the same periodic-FEM homogenisation used to build the catalogue.

7. Discussion

The three regimes above share a single architectural choice – a coordinate-conditioned neural field on top of a differentiable FEM – and they show, in different forms, the same advantage over hand-crafted symmetrisation strategies. In the single-frequency case the network finds a low-distortion realisable cloak by directly searching four orthotropic Cauchy moduli per cell; the broken minor symmetry belongs to the ideal polar tensor and represents the gap that any realisable Cauchy design (ours included) must approximate. In the multifrequency case the network's implicit smoothness eliminates the spurious cell-to-cell oscillations that plague raw per-cell optimisation and the band objective flattens the loss across a wide target band. In the microstructure-constrained case the same network output is projected onto a realisability manifold and realised as an explicit single-material microstructure. This continuum-level performance largely survives realisation: the best-of-N diffusion cloak recovers $\eta = 0.97$ from fabricable single-material cells, with the small residual reflecting per-cell target-matching error rather than a hidden homogenisation gap, since the D_2-invariant cells carry the full orthotropic (4CM) stiffness.

Limitations.. The cell discretisation imposes a piecewise-constant material approximation whose square cells cannot conform exactly to the inclined cloak interfaces; the residual right-boundary distortion of the continuous tensor is dominated by FEM error rather than by an intrinsic limit of the transformation. The microstructure pipeline is currently restricted to a two-phase (single-constituent/void) substrate and a quasi-static homogenised descriptor. The realised cloak is validated at the unit-cell level, since a mesh-converged pixel-resolved solve of the full domain remains computationally out of reach (Sec. 6).

Future work.. The same differentiable pipeline, coupling a neural field to a finite-element solver, applies to other Rayleigh-wave design problems beyond carpet cloaking, such as wave focusing, waveguiding, and vibration isolation, by changing only the objective. On the realizability side, we have already replaced the discrete database with a differentiable generative model of the realisable manifold, queried during optimisation (a guided diffusion model in the spirit of Yang et al. [22]); this model still generates one unit cell at a time, and producing a full multiscale image of the cloak, where neighbouring cells connect and vary smoothly across the domain, is an open problem that may again be a good fit for diffusion models. Such a generator would also relax the square-cell grid used here, which cannot conform to the inclined cloak interfaces: laying cells out on a body-fitted mesh would let the microstructure follow the interfaces and remove a source of residual boundary distortion. Two further extensions stand out. Moving from the two-dimensional carpet cloak to a three-dimensional cloak; the differentiable JAX-FEM pipeline carries over, but the cost of the full solve grows sharply. And designing beyond the four-parameter orthotropic Cauchy class: the ideal polar tensor breaks minor symmetry, so realising it needs Cosserat or polar microstructures and a different homogenisation method than the quasi-static Cauchy one used here.

8. Conclusion

We have presented a unified, differentiable design pipeline for two-dimensional Rayleigh-wave carpet cloaks in which the per-cell material is parameterised by four free orthotropic Cauchy moduli ($C_{11}, C_{12}, C_{22}, C_{66}$) and a scalar density, searched by a coordinate-conditioned neural field through an end-to-end differentiable JAX-FEM pipeline. This is an alternative to closed-form arithmetic-mean symmetrisation of the ideal polar tensor that directly explores the realisable Cauchy design space; the ideal BGM transformation tensor (polar, ten components) is retained as the mathematical reference, while all optimised designs are Cauchy. The pipeline recovers near-perfect single-frequency cloaking, flattens the cloak loss across a target frequency band, and can be projected onto a realisability manifold and realised as explicit binary microstructures built from a single constituent – to our knowledge the first microstructured realisation of a Rayleigh-wave elastodynamic carpet cloak from a common material – recovering approximately 97% of the ideal cloaking performance.

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